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Original article

# Refining the analytic hierarchy process: A statistical and methodological exploration of expert judgments

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ABSTRACT

The research aims to bridge the gap between theoretical assumptions and the actual behavior of experts in real-world scenarios. This gap has significant practical implications, as it affects the validity of the decisions made using AHP. The adjustments proposed are based on a deep statistical and methodological analysis. The analysis of nearly 500 expert judgment matrices reveals that the upper bounds of Saaty's 1–9 scale are rarely used, supporting the adoption of a simplified 1–7 scale to reduce experts' cognitive load and enhance judgment reliability. The authors compared various methods for deriving priority vectors, consistency indices, and deviations. This comparative analysis exposes inconsistencies in traditional approaches and underlines the need for a more context-driven, empirically informed methodology. By decomposing the AHP methodology into eight features with multiple alternatives, our morphological analysis uncovers 4860 potential configurations. This framework not only reveals the latent complexity of AHP but also paves the way for selecting or adapting configurations that better align with the specifics of different decision-making scenarios. The researchers suggest concrete adjustments revising the simulation algorithm for the random consistency index (R.I.) to reconcile theoretical assumptions with empirical behavior.

## 1. Introduction

Decision-making is fundamental to furthering our survival goal and ensuring our quality of life ([16]). [14] highlight that different problems need different solutions, and no one technique can be universally applied to the same effect. Recognized quality-of-life index from the Economist Intelligence Unit, defined for operations with 9 factors. According to [22], many decision-makers still question the usefulness of multi-criteria decision-making (MCDM) methods and prefer to rely on intuitive decisions. [44] state that choosing the MCDM method is a complex problem. [6] emphasize that there is no general conclusion about the "best" MCDM methods. On the other hand, an analysis of the content of a significant number of sources (among many others, [15]; [19]; [31]; [33]) allows us to state that the analytic hierarchy process (AHP) is one of the most widespread MCDM methods. The advantage of the method is the ability to decompose the process under study.

Developed in the 1980s by Saaty, the AHP methodology emerged from an analysis of a limited set of tangible objects. One of Saaty's primary goals was to demonstrate the possibility of relative measurement of intangible objects by comparing them with tangible ones. According to the AHP, a matrix of pairwise comparisons is formed based on experts' judgments. A scale of 1–9 is recommended as the fundamental scale, as it is as good as any other scale, but it has the advantage of simplicity and is, therefore, quite natural ([34]). One key theoretical issue is the intensity scale used in pairwise comparisons. Although [34] originally proposed a 1–9 scale, empirical analyses have shown that experts rarely approach the upper limits of this range. In fact, the more in-depth discussion by [18] illustrates how expert judgments might be better represented by alternative scales—for instance, a 1–7 scale, particularly when dealing with smaller matrices. This insight forms a major motivation for our investigation. Based on the matrix of pairwise comparisons, the authors calculated the leading indices: the priority vector$\overline{w}$, the maximum eigenvalue $\lambda_{max}$, the consistency index C.I., and the consistency ratio C.R.

In the studies using AHP, positive evaluations of the possibilities of this method dominate. [20] argue that AHP has broken through the academic community to be widely used by practitioners. [19] state that there are 19 application areas of AHP, among which manufacturing has attracted the most attention, followed by logistics. Logistics problems are multidimensional and include not only transportation parameters

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but also warehousing, distribution and maintaining inventory levels. [23] indicate that researchers widely use AHP in strategy-related studies. Considerably less attention is paid to the AHP methodology. [40] distinguished AHP applications in operations management (OM) into two categories: papers that suggest a new modeling development (only 20 % of AHP applications) and papers that apply existing AHP-based methodology to real-world problems (nearly 80 % of total AHP applications in OM).

[20] emphasize that choosing the "best'' scale is a heated debate. According to [25], more than 20 methods are available for deriving the priority weights in a pairwise comparison matrix (PCM). According to [27], the choice of algorithm for the ranking is still the subject of research and vigorous debate. [8] notes that not all the facets of pairwise comparison matrices can be fruitfully analyzed using randomly generated matrices. This suggests increasing the use of empirical preferences in studying preference relations. [10] state that Fuzzy AHP is not a favorable method over classical AHP. Also, the relevance of solving the set tasks is caused by the possibility of obtaining the most relevant and reliable parameters by methods of Big Data technologies.

The theoretical foundation of this study is grounded in the principles of ratio scale estimation and the inherent assumptions of the Analytic Hierarchy Process (AHP). One central focus is the intensity scale used for expert judgments. While [34,35] originally proposed a 1–9 scale for pairwise comparisons, empirical evidence and further theoretical insights—such as those provided by [18]—suggest that experts tend to use a narrower range of values, prompting consideration of an alternative 1–7 scale. This study not only re-evaluates the appropriateness of these scales in capturing expert judgment but also establishes a framework for examining various computational methods for deriving priority vectors and consistency indices. By dissecting AHP into many features and alternatives through morphological analysis, we can systematically explore over 4860 potential configurations, thereby linking theoretical postulates with empirical observations. This integrated approach provides the necessary backdrop for assessing how variations in scale and computational techniques might influence the reliability of decisions in multi-criteria contexts.

However, despite the research results aimed at developing the AHP methodology (for example, [30]), the approaches proposed by [34] are still being used to solve practical problems. An overview of several papers is presented in Table 1.

Thus, AHP is one of the most widespread MCDM methods proposed by Saaty in the 1980s based on the study of a limited number of tangible objects. Over the last 40 years, several studies have aimed at developing the AHP methodology. At the same time, works related to practical problems use the fundamental 1–9 scale, and eigenvector method to derive priorities from comparison matrices, calculate the maximum eigenvalue $\lambda_{max}$, generate random variables to estimate the mean random consistency index R.I. (uniform distribution of the values 1/9, 1/8, …, 1/2, 1, 2, …, 9 is simulated), and C.R.$\leq$ 0.1 ([34]). On the other hand, [26] highlight that any flaw in the AHP methodology may lead to incorrect decisions with tremendous negative consequences.

Based on the analysis of expert judgment arrays, our research aims to verify that the AHP methodology proposed in the 1980s can be used to make unambiguous (singularly correct) decisions for real-world problems.

The structure of the paper is as follows. Section 2 summarizes the research of the AHP based on morphological analysis and synthesis. Section 3 compares the scales for assessing experts' judgments based on a statistical analysis of numbers used by experts when filling pairwise comparison matrices. Section 4 gives a comparative analysis of various methods to derive the priority vector, considering the computation method of C.I. and the index of mean values of absolute deviations of expert and actual priority vectors (based on computations for tangible and intangible objects). To test the proposed approaches, an example of estimating total supply chain cost components is given in Section 5. The implications of the study and the direction for future research are offered in Section 6.

**Table 1**
Analysis of using the AHP methodology in modern research.

| Reference | The main purpose of the study | Remarks |
|---|---|---|
| [9] | A heuristic approach that auto-generates a consistent matrix based on the original inconsistent matrix is presented in this paper. | The eigenvector method is used to generate the priority vector. If CR> 0.1, the decision-maker is asked to revise his judgments until an acceptable level of consistency is reached. |
| [32] | A hierarchical weapon systems valuation model framework is discussed. | The 1–9 scale developed by Saaty is used. All the judgments are found at the limit of consistency with a consistency ratio, CR < 0.1. |
| [42] | This paper proposes a novel decentralized dispatch strategy based on the AHP. | Ratio scales demonstrate the relative importance, from overwhelmingly less important (i.e., 1/9) to overwhelmingly more important (i.e., 9. If the consistency ratio is no larger than 10 %, the inconsistency of judgments is acceptable. Otherwise, the judgments need to be revised. |
| [21] | This paper has addressed the ABC inventory classification. | The criteria are pairwise compared in a 1–9 scale matrix. A consistency ratio exceeding 10 % may indicate a set of judgments that are too inconsistent to be reliable, and therefore, it is recommended that the evaluations be revised. |
| [43] | This research intends to discover a method that derives an acceptable consistency PCM | Before a PCM can be used, it must pass a consistency test (acceptable consistency CR<0.1). |
| [17] | This study aims to identify the best cryptocurrency mining strategy. | The normalized pairwise matrix was computed, where all the column elements are divided by the sum of the column. Then, the criteria weights were computed by taking the average of each row. The CI is equal to 0.12, and the C.R. is less than 0.1, the upper bound limit for C.R. acceptance. |

## 2. Methodology

Analytic Hierarchy Process is primarily based on mathematical tools. The morphological analysis and synthesis method is one possible way of summarizing the AHP studies. Based on a morphological analysis and synthesis, it is possible to study the maximum quantity of options of decisions of problems possible within the AHP methodology, which will, in turn, allow options to be considered that might be simplified in the conventional approaches. Such use is particularly in demand in the transitional period of the symbiosis between artificial intelligence and the decision maker.

Morphological analysis is a systematic approach that decomposes a complex decision-making problem into its fundamental characteristics, or dimensions, which are considered to be independent from one another. In our study, this process facilitates the disaggregation of the various methodological "alternatives" within the AHP framework into discrete, mutually exclusive features. By doing so, we can generate a comprehensive set of potential configurations—in this case, over 4860 possible alternatives—allowing for a more detailed and objective comparison of each constituent element. This systematic decomposition not only simplifies the decision process by isolating specific operational choices, but it also maximizes objectivity by ensuring that each dimension is scrutinized independently. Ultimately, the morphological analysis provides a clear and structured framework that enhances the transparency and rigor of our comparative assessments, making it more natural for readers to understand the methodology before delving into

the details of the subsequent analyses.

### 2.1. Morphological analysis of alternative AHP options

The morphological analysis divides the investigated problem into features $H_i$ and alternatives $Y_{ij}$ (Table 2).

In this framework, we decompose the overall AHP methodology into a set of fundamental dimensions, which we refer to as **features**. Each feature represents a distinct aspect or component of the decision-making process within AHP, such as the choice of the scale for expert judgments or the method for deriving the priority vector.

For every feature, there exist multiple **alternatives**—that is, different methods or approaches available for addressing that particular aspect. These alternatives, sometimes also referred to as **options**, represent the specific procedural choices or computational techniques that can be applied in the context of that feature. For example, within the **scale feature**, the alternatives might include the traditional 1–9 scale, a simplified 1–7 scale, or other specialized scaling arrangements.

**Table 2**
Basic morphological features and alternatives of the AHP.

| Feature $H_i$ | Alternatives $Y_{ij}$ of the implementation of the feature $H_i$ | Number of alternatives $n_i$ of the feature $H_i$ |
|---|---|---|
| Scale ($H_1$) | $Y_{11}$ – fundamental scale 1–9 ([34,35]);<br>$Y_{12}$ – scale 1–7;<br>$Y_{13}$ – other (specialized) options, such as geometric ([29]), balanced ([37]) | 3 |
| Object being assessed ($H_2$) | $Y_{21}$ – tangible objects;<br>$Y_{22}$ – intangible objects;<br>$Y_{23}$ – sample including tangible and intangible objects | 3 |
| Methods for deriving the priority vector $w$ ($H_3$) | $Y_{31}$ – summation by rows;<br>$Y_{32}$ – summation by columns;<br>$Y_{33}$ – summation by columns and rows (double);<br>$Y_{34}$ – geometric mean by rows ([13]);<br>$Y_{35}$ – other combined options, such as linear programming method ([11]), Bayesian prioritization procedure ([3]) | 5 |
| Options of calculating the maximum eigenvalue $\lambda_{max}$ and consistency index C.I. ($H_4$) | $Y_{41}$ – classical option ([34]);<br>$Y_{42}$ – other options ([41]), and also, for instance, the geometric consistency index ([1]);<br>harmonic consistency index ([39]) | 2 |
| Methods of calculating the random consistency index R.I. ($H_5$) | $Y_{51}$ – based on the generation of a uniform distributed random value from 1/9–9;<br>$Y_{52}$ – other options | 2 |
| Choice of criterion z's value for assessing the consistency ratio C.R.=C.I./R.I. ($H_6$) | $Y_{61}$ – basic, universal $z = 0.1$ ([34]);<br>$Y_{62}$ – differentiated: $z$ from 0.05 to 0.20 ([2,35]);<br>$Y_{63}$ – other options, such as $z = 0.3$ | 3 |
| Assessment of correctness of the results at C.R.$\leq z$ ($H_7$) | $Y_{71}$ – based on C.I. and C.R. (for all objects being assessed $H_2$);<br>$Y_{72}$ – based on comparing priority vectors (for objects $Y_{22}$);<br>$Y_{73}$ – combined assessments (for objects $Y_{23}$) | 3 |
| Revision of the values of the elements $a_{ij}$ of the pairwise comparison matrix at C.R.$>z$ ($H_8$) | $Y_{81}$ – choice of $a_{ij}$ based on the ratio $\max[a_{ij} - (w_i/w_j)]$;<br>$Y_{82}$ – choice of $a_{ij}$ based on the ratio $\max[a_{ij}/(w_i/w_j)]$;<br>$Y_{83}$ – other options of the choice of $a_{ij}$ ([43]) | 3 |

Similarly, for the feature corresponding to priority vector derivation, the alternatives could cover methods like the arithmetic summation by rows, arithmetic summation by columns, geometric mean, or even methods proposed by other researchers.

This systematic decomposition, where each feature is explored independently and in terms of its alternatives, enables a comprehensive comparison of over 4860 AHP configurations. In doing so, it not only highlights the individual merits and shortcomings of each alternative but also serves to increase the clarity and objectivity of the overall decision-making process.

The formula calculates the number of options N forming a morphological set

$$N = \prod_{i=1}^{L} n_i = n_1 \cdot n_2 \cdot \ldots \cdot n_L, \tag{1}$$

where $n_i$ is the number of alternatives of implementing the $i$-th feature; $L$ is the number of features.

Regarding the calculation $\lambda_{max}$, the researchers consider the availability of various approaches, but most often, the two options occur. The first (classical) option ([34]) includes three operations:

- multiplication of the reciprocal matrix $|A|^{-1}$ by the priority vector $\overline{w} = (w_1, w_2, \ldots, w_n)$ and obtaining the column vector $\overline{Q}$ (i.e. $\overline{Q} = |A| \times \overline{w}$);
- the division of the elements $q_i$ of the column vector $\boldsymbol{Q}$ by the respective elements $w_i$ of the priority vector $\overline{w}$ (i.e. $q_i/w_i$);
- calculation of the maximum eigenvalue $\lambda_{max}$ according to the formula

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^{n} (q_i/w_i). \tag{2}$$

Then, based on $\lambda_{max}$ C.I. is calculated

$$C.I. = \frac{\lambda_{\max} - n}{n - 1}. \tag{3}$$

The second option ([41]) implies the performance of the first operation ($\boldsymbol{A} \cdot \boldsymbol{w} = \boldsymbol{Q}$), then the maximum eigenvalue is calculated according to the formula

$$\lambda_{\max} = \sum_{i=1}^{n} q_i \tag{4}$$

Thus, according to Table 2, the number of options for forming AHP is Nmax= 4860 (without considering the alternatives' interrelation). For practical application of the presented calculations it is enough to use the capabilities of the widespread MS Office package, where all the necessary operations are realized, namely add-ons for Microsoft Excel: Solver (An add-in for Microsoft Excel that performs optimization by finding the maximum or minimum value for a formula subject to constraints), MMULT (The function «matrix multiplication» multiplies two matrices and returns the matrix product), SUMPRODUCT (The function «sum product» multiplies corresponding components in given arrays and returns the sum of those products), MINVERSE (The function «matrix inverse» calculates the inverse of a given matrix).

### 2.2. Morphological synthesis of alternative AHP options

Similar to the approach to the designation of queuing systems suggested in the research ([41]), it is proposed to submit a basic option of computational operations conducted in the AHP using a structural model of a multicomponent vector of alternatives for the option developed by [34] is written as

$$S(Y_{ij}) = (Y_{11}/Y_{23}/Y_{33}/Y_{41}/Y_{51}/Y_{61}/Y_{71}/Y_{82}), \tag{5}$$

where $Y_{ij}$ is respectively the $j$-th alternative of implementing the $H_i$ feature.

The authors suppose that in this form, the AHP model does not consider the specifics of the $H_7$ feature (assessment of the correctness of the result), reflecting the availability of feedback and the possibility of correcting the matrix of experts' judgments. If the computational value of C.R. exceeds (Table 1) the admissible criterion $z$ (feature $H_6$), a transition to the procedure of revision of the original matrix (feature $H_8$) is required, i.e. the re-calculation, starting from the $H_3$ feature. We should repeat this procedure is until the condition C.R.$\leq z$ is fulfilled.

The structural AHP model, formula (5), and the unified option can be represented as interrelated models. The first model includes features allowing the primary indices to be calculated (priority vector $w$, maximum eigenvalue $\lambda_{max}$, consistency index C.I.) and can be represented as

$$S_1(H_i) = (H_1/H_2/H_3/H_4). \tag{6}$$

Eq. (6) represents the comprehensive model for computing the priority vector $\overline{w}$, the maximum eigenvalue $\lambda_{\max}$, and the consistency index *C.I.* It integrates the computations of Eqs. (2) and (3) into a single framework. This model is foundational for determining whether the expert judgments in the AHP matrix are sufficiently consistent.

The second model, including control and adjusting operations, is written as

$$S_2(H_i) = (H_5/H_6/H_7/H_8). \tag{7}$$

Eq. (7) provides an additional layer of validation beyond the initial computations. Based on the conducted analysis for further development of AHP, additional investigations will be required initially related to the first model, formula (6), as it is the model $S_1(H_i)$ that will be the basis for the model $S_2(H_i)$ during the transition from the $H_5$ feature to the $H_6$ feature.

## 3. Choice of scales for assessment of experts' judgments

Let us discuss the first structural model, formula (6). Based on analyzing studies by Saaty ([34,35]) that are the basis for forming AHP, we assess matrices reflecting experts' judgments and specific results of their processing.

Table 3 systematizes data on the total number of matrices of various dimensions (from 3 × 3–8 × 8) considering the specifics of tangible or intangible objects of assessment. Table 4 shows each matrix's maximum values of experts' judgments and the number of consistent and inconsistent matrices.

It follows from analyzing Tables 3 and 4 that:

- of 109 matrices, almost 84 % are 3 × 3 and 4 × 4, whereas the share of 7 × 7 and 8 × 8 matrices is below 4 %;
- only 6.4 % of matrices present information on the assessment of tangible objects (alternative $Y_{21}$); the remaining 93.6 % are matrices of intangible objects. Thus, the ratio between the two types of matrices is 1:15;
- a comparison of maximum values of experts' judgments for various matrices shows that scales 1–9, 1–7, and 1–5 are used with almost the same frequency, but a certain regularity is observed: for 3 × 3 and 4 × 4 matrices, experts prefer scales 1–7 (25.3 %) and 1–5 (34.6);

**Table 4**
Distribution of maximum values of experts' judgments and assessments of consistency of matrices (%) ([34,35]).

| Feature ($H_i$) | Maximum judgment | Matrices | | | | | Total |
|---|---|---|---|---|---|---|---|
| | | 3 × 3 | 4 × 4 | 5 × 5 | 6 × 6 | 7 × 7 (8 × 8) | |
| Scale ($H_1$) | 9 or 8 | 20.3 | 3.8 | 2.0 | 1.9 | 2.9 | 30.9 |
| | 7 or 6 | 12.3 | 13.0 | 0.9 | 6.5 | 0.9 | 33.6 |
| | 5 and below | 24.4 | 10.2 | 0.9 | - | - | 35.5 |
| | total | 57.0 | 27.0 | 3.8 | 8.4 | 3.8 | 100 |
| Assessment of consistency ($H_7$) | consistent (C.R.$\leq z$) | 45.7 | 6.2 | 3.7 | 2.5 | 3.7 | 61.8 |
| | inconsistent (C.R.$>z$) | 22.2 | 8.4 | 1.2 | 4.9 | 1.5 | 38.2 |
| | result | 67.9 | 14.6 | 4.9 | 7.4 | 5.2 | 100 |

- 62 % of matrices are consistent for an analyzed sample, with the value for 7 × 7 and 8 × 8 matrices reaching 75 %.

Thus, the results presented in Tables 3 and 4 contradict one of Saaty's primary goals: to demonstrate the possibility of relative measurement of intangible objects by comparing them with tangible ones.

To support the legitimacy of the statement on the efficiency of the scale 1–9, the research by [34] presents the results of calculations for three tangible objects of assessment: brightness of chairs (4 × 4 matrices), transportation distance (6 × 6 matrices), and national wealth of several countries (7 × 7 matrices). The obtained actual priority values are summarized in a set of $\overline{w}_s = (w_{s1}, w_{s2}, \ldots, w_{sn})$. One expert formed each matrix, and $\lambda_{max}$ and standard deviation $\sigma$ were calculated for each of them

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(w_i - w_{si})^2}, \tag{8}$$

where $w_i$, $w_{si}$ are the calculated (expert estimates) and actual values of the $i$-th of a priority, respectively.

To prove the unambiguity of the choice of the scale 1–9, calculations for 27 scales were performed ([34]). The analysis of these calculations showed that the close results are observed for two scales: 1–7 and 1–9. Therefore, as an alternative, the scale 1–7 was chosen.

Table 5 presents the results of calculations of matrices for tangible objects conducted according to the morphological features (Table 2): scales 1–9 and 1–7 ($Y_{11}$ and $Y_{12}$); tangible objects ($Y_{21}$); summation by columns and rows to derive the priorities $w_i$ ($Y_{33}$); classical option of calculation $\lambda_{max}$ and C.I. ($Y_{41}$).

Along with the calculations of $\lambda_{max}$ and $\sigma$ performed by Saaty, we calculated C.I. and mean values of absolute deviation:

$$\Delta = \frac{1}{n}\sum_{i=1}^{n}|w_i - w_{si}| \tag{9}$$

Table 5 was supplemented by three more matrices: 3 × 3, 5 × 5, and 8 × 8. The 3 × 3 matrix was formed by an expert according to the assessment results of lengths of three rods of equal weight and diameter and made of bronze, textile laminate, and wood. The 5 × 5 matrix is a

**Table 3**
Distribution of matrices accounting for the specifics of the tangible and intangible objects of assessment ([34,35]).

| Object of assessment ($H_2$) | Matrices | | | | | | Total | |
|---|---|---|---|---|---|---|---|---|
| | 3 × 3 | 4 × 4 | 5 × 5 | 6 × 6 | 7 × 7 | 8 × 8 | Number of matrices | In percent |
| Intangible ($Y_{21}$) | 65 | 24 | 4 | 8 | 0 | 1 | 102 | 93.6 |
| Tangible ($Y_{22}$) | 0 | 2 | 1 | 1 | 2 | 1 | 7 | 6.4 |
| Total | 65 | 26 | 5 | 9 | 2 | 2 | 109 | - |
| Share in sample (%) | 59.7 | 23.8 | 4.6 | 8.3 | 1.8 | 1.8 | - | 100 |

**Table 5**
Results of calculation of consistency indices of matrices for tangible objects for the scales 1–7 and 1–9.

| Object | Matrix | Scale | $\lambda_{max}$ | $\sigma$ | C.I. | $\Delta$ |
|---|---|---|---|---|---|---|
| Rods of various materials | 3 × 3 | 1–7 | 3.002* | 0.031 | 0.001* | 0.028 |
| | | 1–9 | 3.027 | 0.020* | 0.014 | 0.019* |
| Brightness of chairs | 4 × 4 | 1–7 | 4.034* | 0.019 | 0.012* | 0.016 |
| | | 1–9 | 4.102 | 0.008* | 0.034 | 0.007* |
| Geometric shapes | 5 × 5 | 1–7 | 5.131 | 0.020 | 0.033 | 0.015 |
| | | 1–9 | 5.024* | 0.014* | 0.006* | 0.010* |
| Distance of air flights | 6 × 6 | 1–7 | 6.190* | 0.019 * * | 0.038* | 0.021 |
| | | 1–9 | 6.454 | 0.019 * * | 0.091 | 0.017* |
| Assessment of the wealth of the nations | 7 × 7 | 1–7 | 7.305* | 0.017 | 0.051* | 0.013 * * |
| | | 1–9 | 7.608 | 0.014* | 0.101 | 0.012 * * |
| Power consumption by household electric appliances | 8 × 8 | 1–7 | 8.945* | 0.023* | 0.135* | 0.022 * * |
| | | 1–9 | 9.256 | 0.027 | 0.180 | 0.021 * * |

* *) same (or close) values of indices of one matrix
* ) most negligible value of the index for the respective matrix

known problem of the areas of five geometric shapes ([36]), and the 8 × 8 matrix is a comparison of an assessment of annual power consumption by household electric appliances ([34]).

It follows from the analysis of Table 5 that:

- four indices can be combined into two groups: the first one includes $\lambda_{max}$ and C.I. and reflects matrix consistency; the second one contains indices $\sigma$ and $\Delta$ reflecting the deviation of the values of experts' judgments from the actual values of assessments of objects;
- indices of the first group ($\lambda_{max}$ and C.I.) for five objects out of six have lesser values for the scale 1–7 compared to the scale 1–9;
- indices of the second group ($\sigma$ and $\Delta$) have lesser values in using the scale 1–9; however, it should be noted that for intangible objects, the calculation $\sigma$ and $\Delta$ is impossible;
- only for the 5 × 5 matrix did all indices turn out to be smaller using the scale 1–9.

Thus, despite the limited number of cases of tangible objects, the statement of the overwhelming superiority of the scale 1–9 over other scales is debatable and requires further investigation.

As an alternative option of assessment of the unambiguity of the scale 1–9 applied in AHP, let us review the approach based on an analysis of pairwise comparison matrices from the sources presented in the Introduction section (International Journal of Production Research, International Journal of Production Economics, European Journal of Operational Research, and others), as well as based on the results of our research.

Considering the procedure of filling reciprocal matrices, the numbers in the upper triangular section (above the main diagonal) of the matrix were used for analysis. From these, three samples were formed. The first sample included the numbers reflecting the experts' judgments $a_{ij}$ in the pairwise comparison matrix, being subject to the ratio $1 \le a_{ij} \le 9$;; the second sample included the values of the denominators of fractions from 1/9–1/2 ($1/9 \le a_{ij} \le 1/2$), i.e., 9, 8, …, 2. The third sample is the sum of the first and second samples and characterizes the statistics of the fundamental scale 1–9 for various matrices. For instance, for the 5 × 5 matrices, the first sample will include six numbers: 9, 2, 4, 7, 3, 4; the second sample, numbers 6, 2, 3, 5; and the third one, numbers 9, 2, 4, 7, 3, 4, 6, 2, 3, 5.

$$A = \begin{bmatrix} 1 & 9 & 2 & 4 & 7 \\ 1/9 & 1 & 1/6 & 1/2 & 1/3 \\ 1/2 & 6 & 1 & 3 & 1/5 \\ 1/4 & 2 & 1/3 & 1 & 4 \\ 1/7 & 3 & 5 & 1/4 & 1 \end{bmatrix}$$

[4] presented eight 4 × 4 matrices with 48 experts' judgments. We calculated the mean values $\overline{x}$ and standard deviation $\sigma$ for each sample (Table 6). The numbers 8 and 9 were used four times (the empirical probability was 8.4 %).

In our empirical analysis of expert judgment matrices, we validated the consistency and distribution of numerical assessments by analyzing the numbers from the upper triangular sections. Table 3 systematizes the distribution of matrices by dimensions and the corresponding types of objects assessed. Table 4 then zooms in on the maximum values of expert judgments and the frequency of consistent versus inconsistent matrices. In particular, Table 5 compares the consistency indices ($\lambda_{\max}$ and C.I.) and deviation measures ($\sigma$ and $\Delta$) between the scales 1–7 and 1–9, demonstrating that for 5 × 5 matrices the overall inconsistency is lower when using the scale 1–7. Furthermore, Table 6 provides detailed statistics on the numbers featured in the upper triangular section of the matrices, reporting the quantity, standard deviation, and the percentage share of expert judgments for each number. This quantitative analysis is further complemented by the aggregated data presented in Tables 7 and 8, which summarize statistical parameters and probabilities across all 472 matrices. The combined data from these tables clearly illuminate the patterns in expert judgments and bolster the argument for adopting a 1–7 scale to reduce cognitive load and maintain consistency.

Tables 7 - 8 present systematized data for the three types of samples obtained based on an analysis of 472 matrices. For instance, 123 5 × 5 matrices were analysed. The number 1 in the experts' judgments was featured 60 times, the number 2 was used 209 times, the number 1/9 was used by the experts 33 times, etc. A total of 1230 judgments were given by the experts, which were in the upper triangular section of the matrix, and of these, 810 (i.e. 66 %) included numbers from 1 to 9 (first sample) and 420 (i.e. 34 %) from 1/9–1/2. For the first sample the mean value was $\overline{x} = 3.75$, standard deviation was $\sigma = 2.66$. The number 8 was used by the experts in the pairwise comparison matrix 78 times, therefore the empirical probability of getting into the samples was 6.3 %; numbers 8 and 9 were used 140 times, and the probability was 11.3 %. Thus, for 123 pairwise comparison 5 × 5 matrices, the mean value of the experts' judgments and the standard deviation $\sigma = 2.25$.

The following conclusions can be drawn based on the analysis of Tables 7 and 8.

1. The mean values $\overline{x}$ for all samples of the matrices (from 3 × 3–7 × 7) are pretty narrow, from 3.72 to 4.81.

2. There are stable ratios between the sizes of the first- and second-type samples: the size of the first-type samples ranges from 63 % to 74 %, while the size of the second-type samples ranges from 26 % to 37 %.

3. The empirical probabilities reflecting experts' use of numbers when filling in the pairwise comparison matrix were from 4.1 % to 10 % (for the number 8) and from 4.8 % to 6.2 % (for the number 9). The empirical probabilities $P(8+9)$ characterizing the use of numbers 8 and 9 in the third sample were from 10.2 % (for the 3 × 3 matrices) to 15.5 % (7 × 7 matrices).

4. The obtained data contradict the basic hypothesis used for the generation of random values for the assessment of a mean random consistency index R.I. According to this hypothesis, a uniform

**Table 6**
Statistical data on the quantity of numbers in the upper triangular section of the matrices.

| Matrix | Sample | Quantity of experts' judgments in pairwise comparison matrices | | | | | | | | | Quantity of judgments $a_{ij}$ | $\bar{x}$ | $\sigma$ | Share of judgments in sample 3 (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | | | | |
| 4 × 4 | 1 | 3 | 3 | 10 | 5 | 4 | 0 | 1 | 2 | 1 | 29 | 3.83 | 2.01 | 60 |
| | 2 | - | 3 | 8 | 1 | 4 | 2 | 0 | 0 | 1 | 19 | 3.95 | 1.73 | 40 |
| | 3 | 3 | 6 | 18 | 6 | 8 | 2 | 1 | 2 | 2 | 48 | 3.88 | 1.91 | - |

**Table 7**
Statistical data on the numbers in the upper triangular section of the 472 matrices.

| Matrix | Sample | Quantity of experts' judgments in pairwise comparison matrices | | | | | | | | | Quantity of judgments $a_{ij}$ | $\bar{x}$ | $\sigma$ | Share of judgments in sample 3 (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | | | | |
| 3 × 3 | 1 | 18 | 31 | 44 | 26 | 40 | 20 | 18 | 13 | 16 | 226 | 4.46 | 2.40 | 71 |
| | 2 | - | 23 | 36 | 8 | 12 | 2 | 7 | - | 4 | 92 | 3.72 | 1.87 | 29 |
| | 3 | 18 | 54 | 80 | 34 | 52 | 22 | 25 | 13 | 20 | 318 | 4.25 | 2.19 | - |
| 4 × 4 | 1 | 33 | 131 | 126 | 92 | 64 | 56 | 41 | 25 | 38 | 606 | 4.15 | 2.21 | 74 |
| | 2 | - | 35 | 46 | 24 | 57 | 13 | 14 | 8 | 13 | 210 | 4.50 | 2.64 | 26 |
| | 3 | 33 | 166 | 172 | 116 | 121 | 69 | 55 | 33 | 51 | 816 | 4.24 | 2.16 | - |
| 5 × 5 | 1 | 60 | 209 | 158 | 92 | 84 | 68 | 51 | 49 | 29 | 810 | 3.75 | 2.66 | 66 |
| | 2 | - | 91 | 77 | 73 | 61 | 32 | 24 | 29 | 33 | 420 | 4.52 | 2.18 | 34 |
| | 3 | 60 | 300 | 235 | 165 | 145 | 100 | 75 | 78 | 62 | 1230 | 4.18 | 2.25 | - |
| 6 × 6 | 1 | 75 | 165 | 150 | 116 | 107 | 73 | 54 | 60 | 35 | 835 | 4.14 | 1.86 | 63 |
| | 2 | - | 90 | 89 | 71 | 76 | 57 | 41 | 34 | 29 | 485 | 4.70 | 2.04 | 37 |
| | 3 | 75 | 255 | 239 | 187 | 183 | 130 | 95 | 94 | 64 | 1320 | 4.34 | 2.20 | - |
| 7 × 7 | 1 | 29 | 30 | 42 | 35 | 33 | 33 | 26 | 32 | 19 | 279 | 4.76 | 2.42 | 70 |
| | 2 | - | 13 | 24 | 17 | 25 | 15 | 15 | 8 | 3 | 120 | 4.81 | 1.87 | 30 |
| | 3 | 29 | 43 | 66 | 52 | 58 | 48 | 41 | 40 | 22 | 399 | 4.77 | 2.29 | - |

**Table 8**
Statistical parameters of numerical judgments used by experts for pairwise comparison of the 472 matrices of various dimensions.

| Matrix | Quantity | | Statistical assessments | | Probabilities $P$ of getting into samples of numbers 8; 9; 8 and 9 (%) | | |
|---|---|---|---|---|---|---|---|
| | of matrices | of judgments | $\bar{x}$ | $\sigma$ | $P(8)$ | $P(9)$ | $P(8+9)$ |
| 3 × 3 | 106 | 318 | 4.25 | 2.19 | 4.1 | 6.2 | 10.3 |
| 4 × 4 | 136 | 816 | 4.24 | 2.16 | 4.1 | 6.1 | 10.2 |
| 5 × 5 | 123 | 1230 | 4.18 | 2.25 | 6.3 | 5.0 | 11.3 |
| 6 × 6 | 88 | 1320 | 4.34 | 2.20 | 7.1 | 4.8 | 11.9 |
| 7 × 7 | 19 | 399 | 4.77 | 2.29 | 10.0 | 5.5 | 15.5 |

distribution of the values 1/9, 1/8, …, 1/2, 1, 2, …, 9 is simulated, and the probability of using the numbers 8 and 9 $P(8+9)$ should amount to 23.5 %. Thus, the simulation algorithm R.I. should be revised.

5. For the 3 × 3, 4 × 4 and 5 × 5 matrices, using the scale 1–7 is appropriate.

Thus, the results show that experts in filling pairwise comparison matrices based on the fundamental scale 1–9 use numbers 8 and 9 much more rarely than implied in the AHP methodology, i.e., they use the scale 1–7. Thus, the simulation algorithm R.I. (feature $H_5$ of the morphological table) should be revised. The results are significant: the R.I. value is used to calculate the consistency ratio C.R.=C.I./R.I., with most studies (such as those shown in Table 1) indicating the need to revise expert judgments when C.R.> 0.1.

It has been established that in using the scale 1–7, the indices $\lambda_{max}$, C.I., $\Delta$, and $\sigma$ calculated for tangible objects are not worse than those of the said indices in using the scale 1–9. Moreover, using the scale 1–7 allows experts' cognitive effort to be reduced in connection with a decrease of the numbers by 22 % compared to the scale 1–9 (13 numbers instead of 17).

**Table 9**
Four methods to derive priority vectors and formulas for mean values.

| Method and computational formula | Description | Computational formulas for mean values |
|---|---|---|
| First ($Y_{31}$ – summation by rows) $w_i = \sum_{j=1}^{n} a_{ij} / \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}$ | Sum the values $a_{ij}$ of each $i$-th row and normalize by dividing the sums in each row by the sum of all elements. | Mean arithmetic $A(x) = \left(\sum_{i=1}^{n} x_i\right)/n$ |
| Second ($Y_{32}$ – summation by columns) $w_i = \frac{1}{\sum_{j=1}^{n} a_{ij}} / \sum_{j=1}^{n} \frac{1}{\sum_{j=1}^{n} a_{ij}}$ | Sum the values of elements of each column ($j$) and obtain the reciprocal values of these sums; normalize them so that the sum equals one. | Mean harmonic $H(x) = n/\left(\sum_{i=1}^{n} 1/x_i\right)$ |
| Third ($Y_{33}$ – summation by columns and rows (double) $w_i = \frac{1}{n} \sum_{i=1}^{n} \left(a_{ij} / \sum_{j=1}^{n} a_{ij}\right)$ | Divide the elements of each column by their sum (by column), then add up the normalized elements by row and divide the sum by row by $n$. | - |
| Fourth ($Y_{34}$ – geometric mean by rows) $w_i = \frac{\left(\prod_{j=1}^{n} a_{ij}\right)^{1/n}}{\sum_{i=1}^{n} \left(\prod_{i=1}^{n} a_{ij}\right)^{1/n}}$ | Multiply the values $n$ of elements of each row and derive the root of the $n$-th degree; normalize the obtained values (by vertical). | Mean geometric $G(x) = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$ |

$a_{ij}$ – expert judgment ($i$-th row, $j$-th column); $n$ – matrix dimension

## 4. Analysis of the methods and options to derive the priority vector $w$, the maximum Eigenvalue of the matrix $\lambda_{max}$ and consistency index C.I

According to [34], the four methods to derive priorities from comparison matrices (Table 9) deserve the most significant attention, as it is asserted that the accuracy of the calculation grows from the first ($Y_{31}$) to the second ($Y_{32}$) and then to the third ($Y_{33}$) method, but the excellent approximation can be obtained only with the fourth method ($Y_{34}$).

However, it is known that the mean arithmetic value *A(x)* exceeds the mean geometric value *G(x)*, which in turn exceeds the mean harmonic *H(x)*:

$$A(x) > G(x) > H(x). \tag{10}$$

For instance, for the numbers 1, 2 and 4 we find:$A(x) = 2.3$; $G(x) = \sqrt[3]{1 \cdot 2 \cdot 4} = 2$;$H(x) = 3/(1 + 0.5 + 0.33) = 1.62$.

Thus, with compliance of the computational formulas for mean values with the Saaty methods ([34]), the minor assessments will correspond to the second, fourth, and most significant assessments to the first. Regarding the third method, there are no unambiguous answers here, and the results of the respective calculations can be used to conduct its assessment.

Summarizing the conducted analysis of the $H_3$ and $H_4$ features of morphological Table 2, it can be stated that it is about choosing the most verified and valid methods of calculating $w$, $\lambda_{max}$ and C.I., first for the matrices of tangible objects, and second their application for the matrices of intangible objects.

Table 10 presents the results of calculations for five matrices (tangible objects of assessment presented in Table 5), for each of which 12 indices were calculated that can be divided into three groups:

C.I. (1) – four C.I. values were calculated according to the alternative $Y_{41}$ ([34]) based on priority vector methods (alternatives $Y_{31}$ - $Y_{34}$), e.g. for matrix 5 × 5 the C.I. values were 0.135, 0.005, 0.006, and 0.003;

C.I. (2) – four C.I. values were calculated according to the alternative $Y_{42}$ ([41]) based on priority vector methods (alternatives $Y_{31}$ - $Y_{34}$), e.g. for matrix 5 × 5 the C.I. values were 0.193, 0.002, 0.014, and 0.003;

Δ - four indices of mean values of absolute deviations of expert (priorities from comparisons) and actual priority vectors, alternative $Y_{72}$ (for matrix 5 × 5 the Δ values were 0.012, 0.006, 0.010, and 0.014).

Analysis of the obtained results allows the following conclusions to be drawn:

1. If the least values of the indices C.I. and Δ are taken as the best assessment, it can be noted that each analyzed group C.I. (1), C.I. (2) and Δ includes one to four alternatives $Y_{31}$ - $Y_{34}$. For instance, for the group C.I. (1) there are alternatives $Y_{31}$ (4 × 4 matrix), $Y_{33}$ (6 × 6 matrix), and $Y_{34}$ (3 × 3, 5 × 5 and 7 × 7 matrices).

2. Out of fifteen values C.I. (1), C.I. (2) and Δ for each of the alternatives $Y_{31}$ - $Y_{34}$, the five least values refer to the alternative $Y_{34}$ (0.009, 0.003, 0.084, 0.006, 0.014), and four values to the alternative $Y_{32}$ (0.007, 0.029, 0.002, 0.006).

3. In comparing the least values C.I. (1) and C.I. (2), it can be stated that for the group C.I. (2) the least value is observed three times for the alternative $Y_{32}$: 0.007, 0.029, 0.002 for the 3 × 3, 4 × 4 and 5 × 5 matrices, respectively; for the group C.I. (1) the most negligible value is observed three times for the alternative $Y_{34}$: 0.009, 0.003, 0.084 for the 3 × 3, 5 × 5 and 7 × 7 matrices, respectively.

4. In assessing Δ the opposite picture is observed: the least values of this index are observed for the alternative $Y_{34}$ (4 × 4 and 6 × 6 matrices) and once for the alternatives $Y_{31}$, $Y_{32}$ and $Y_{33}$ for the remaining matrices.

It is evident that based on a small group of matrices of tangible objects it is difficult to give preference to one of the options of calculation, and further investigations for the matrices of tangible objects are required.

For the second option of calculation $\lambda_{max}$ and C.I. (alternative $Y_{42}$), we conducted an additional series of experiments, the participants of which were specialists (managers) of companies. The experts individually formed various matrices (of assessment of tangible and intangible objects) based on which $w$, $\lambda_{max}$ and C.I. were calculated. A total of 201 matrices were processed, and the results of the calculations are presented in Table 11. Analyzing Table 11 shows that the least mean values and standard deviation (SD) of the indices $\lambda_{max}$ and C.I. are unambiguously observed for the second method to derive the priority vector (alternative $Y_{32}$).

The continuation of this trend of investigations is the performance of calculations and the respective analysis of matrices published in papers by specialists in various countries. From a library of matrices we collected, 25 of each dimension (from 3 × 3–6 × 6) were taken randomly, the processing results of which are given in Tables 12 and 13.

Despite the small size of the samples, it follows from analyzing Tables 12 and 13 that:

- for the alternative $Y_{41}$ (Table 12) the most negligible value C.I. is observed for the third and fourth methods to derive the priority vector $\overline{w}$;
- for the alternative $Y_{42}$ (Table 13), the least values C.I. are obtained from the second method to derive the priority vector $\overline{w}$.

Thus, it follows from comparing the results of statistical processing of the randomly formed samples that there is no unambiguous decision in choosing a definite method to derive the priority vector $w$ and the option of calculating $\lambda_{max}$ and C.I. At the same time, in comparing the two options of the alternatives $Y_{41}$ and $Y_{42}$, the least values C.I. were obtained for the second method (alternative $Y_{32}$).

The sizes of the samples should be increased to allow for the final decision.

## 5. Case study

Consider an example of estimating the components of total supply chain costs:

$$C_{\sum} = \sum_{i=1}^{n} C_i = C_1 + C_2 + C_3 + C_4, \tag{11}$$

where $C_1$ = ordering costs, $C_2$ = transportation costs, $C_3$ = holding costs (stock on hand), and $C_4$ = holding costs (safety stock). Since the described process is essentially dynamic, we consider the average parameters $C_i$ for $i = 1, 2, 3, 4$ durational.

**Table 10**
Results of processing the matrices (according to alternatives $Y_{31}$ - $Y_{34}$ and $Y_{41}$ - $Y_{42}$).

| Matrix | Option of calculation $\lambda_{max}$ | First method $Y_{31}$ | | Second method $Y_{32}$ | | Third method $Y_{33}$ | | Fourth method $Y_{34}$ | |
|---|---|---|---|---|---|---|---|---|---|
| | | C.I. | Δ | C.I. | Δ | C.I. | Δ | C.I. | Δ |
| 3 × 3 | $Y_{41}$ | 0.017 | 0.013* | 0.014 | 0.047 | 0.014 | 0.019 | 0.009* | 0.034 |
| | $Y_{42}$ | 0.075 | | 0.007* | | 0.013 | | 0.008 | |
| 4 × 4 | $Y_{41}$ | 0.033* | 0.019 | 0.069 | 0.026 | 0.034 | 0.007 | 0.024 | 0.006* |
| | $Y_{42}$ | 0.114 | | 0.029* | | 0.058 | | 0.056 | |
| 5 × 5 | $Y_{41}$ | 0.135 | 0.012 | 0.005 | 0.006* | 0.006 | 0.010 | 0.003* | 0.014 |
| | $Y_{42}$ | 0.193 | | 0.002* | | 0.014 | | 0.003 | |
| 6 × 6 | $Y_{41}$ | 0.127 | 0.018 | 0.337 | 0.044 | 0.091* | 0.017 | 0.105 | 0.014* |
| | $Y_{42}$ | 0.247 | | 0.181 | | 0.091* | | 0.094 | |
| 7 × 7 | $Y_{41}$ | 0.109 | 0.030 | 0.184 | 0.029 | 0.101 | 0.012* | 0.084* | 0.014 |
| | $Y_{42}$ | 0.202 | | 0.065 | | 0.053* | | 0.113 | |

* least value of index

**Table 11**
Results of statistical processing of matrices (according to alternatives $Y_{31}$ - $Y_{34}$ and $Y_{42}$).

| Matrix | Quantity of matrices | Parameter | First method $Y_{31}$ | | Second method $Y_{32}$ | | Third method $Y_{33}$ | | Fourth method $Y_{34}$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. |
| 3 × 3 | 54 | Mean | 3.330 | 0.165 | 3.053 | 0.026 * | 3.211 | 0.106 | 3.132 | 0.066 |
| | | SD | 0.410 | 0.202 | 0.132 | 0.066 * | 0.370 | 0.185 | 0.238 | 0.119 |
| 4 × 4 | 38 | Mean | 4.535 | 0.180 | 4.119 | 0.040 * | 4.356 | 0.120 | 4.238 | 0.080 |
| | | SD | 0.683 | 0.230 | 0.178 | 0.060 * | 0.535 | 0.180 | 0.356 | 0.120 |
| 5 × 5 | 50 | mean | 5.665 | 0.166 | 5.238 | 0.059 * | 5.570 | 0.131 | 5.427 | 0.107 |
| | | SD | 0.665 | 0.166 | 0.285 | 0.083 * | 0.853 | 0.154 | 0.435 | 0.119 |
| 6 × 6 | 59 | mean | 8.112 | 0.422 | 6.990 | 0.198 * | 7.650 | 0.330 | 7.254 | 0.251 |
| | | SD | 1.780 | 0.356 | 0.403 | 0.278 * | 1.584 | 0.317 | 1.320 | 0.264 |

*least value C.I.

**Table 12**
Results of statistical processing of matrices using the alternative $Y_{41}$.

| Matrix | Parameter | First method $Y_{31}$ | | Second method $Y_{32}$ | | Third method $Y_{33}$ | | Fourth method $Y_{34}$ | |
|---|---|---|---|---|---|---|---|---|---|
| | | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. |
| 3 × 3 | mean | 3.44 | 0.22 | 3.43 | 0.22 | 3.36 | 0.18 * | 3.36 | 0.18 * |
| | SD | 0.59 | 0.30 | 0.62 | 0.31 | 0.57 | 0.28 | 0.55 | 0.27 * |
| 4 × 4 | mean | 4.40 | 0.15 | 4.42 | 0.15 | 4.29 | 0.11 | 4.28 | 0.10 * |
| | SD | 0.47 | 0.17 | 0.53 | 0.18 | 0.37 | 0.13 | 0.35 | 0.12 * |
| 5 × 5 | mean | 5.51 | 0.14 | 5.60 | 0.16 | 5.44 | 0.13 * | 5.41 | 0.13 * |
| | SD | 0.43 | 0.12 | 0.55 | 0.14 | 0.39 | 0.11 | 0.36 | 0.10 * |
| 6 × 6 | mean | 7.57 | 0.31 | 7.52 | 0.30 | 7.35 | 0.27 | 7.29 | 0.26 * |
| | SD | 1.86 | 0.37 | 1.83 | 0.36 | 1.85 | 0.37 | 1.83 | 0.36 * |

*least value C.I.

**Table 13**
Results of statistical processing of matrices using the alternative $Y_{42}$.

| Matrix | Parameter | First method $Y_{31}$ | | Second method $Y_{32}$ | | Third method $Y_{33}$ | | Fourth method $Y_{34}$ | |
|---|---|---|---|---|---|---|---|---|---|
| | | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. | $\lambda_{max}$ | C.I. |
| 3 × 3 | mean | 3.53 | 0.27 | 3.24 | 0.12* | 3.45 | 0.22 | 3.35 | 0.18 |
| | SD | 0.56 | 0.28 | 0.50 | 0.25* | 0.61 | 0.31 | 0.55 | 0.27 |
| 4 × 4 | mean | 4.61 | 0.20 | 4.16 | 0.05* | 4.42 | 0.14 | 4.31 | 0.10 |
| | SD | 0.59 | 0.20 | 0.22 | 0.07* | 0.53 | 0.18 | 0.41 | 0.14 |
| 5 × 5 | mean | 5.74 | 0.19 | 5.27 | 0.07* | 5.60 | 0.15 | 5.45 | 0.11 |
| | SD | 0.62 | 0.15 | 0.29 | 0.07* | 0.56 | 0.14 | 0.45 | 0.11 |
| 6 × 6 | mean | 7.93 | 0.39 | 6.88 | 0.18* | 7.51 | 0.30 | 7.15 | 0.23 |
| | SD | 1.88 | 0.38 | 1.66 | 0.33* | 1.84 | 0.37 | 1.66 | 0.33 |

* least value C.I.

A group of five experts was engaged to estimate the costs of $C_i$ in the supply chain.

The first stage was to provide experts with information on the indicators that characterize one of the supply chains presented in the studies of [7], [12], [24], [28], and [38]. In particular: $d$ = demand per time unit; $A$ = ordering and transportation cost per batch; $h$ = holding cost per unit and time unit (stock on hand and safety or emergency stock).

**Table 14**
Results of calculation of C.I. and Δ based on actual data and expert estimates of supply chain costs.

| Reference | $d$ | $A$ | $h$ | $\lambda_{max}$ | Scale 1–7 | | | | Scale 1–9 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Second method $Y_{32}$ | | Third method $Y_{33}$ | | Second method $Y_{32}$ | | Third method $Y_{33}$ | |
| | | | | | C.I. | Δ | C.I. | Δ | C.I. | Δ | C.I. | Δ |
| [12] | 1000 | 100 | 10 | $Y_{41}$ | 0.803 | 0.091 | 0.790* | 0.060 | 1.117 | 0.086 | 1.103 | 0.055* |
| | | | | $Y_{42}$ | 0.852 | | 0.802* | | 1.208 | | 1.121 | |
| [28] | 900 | 50 | 11.2 | $Y_{41}$ | 0.750* | 0.224 | 0.814 | 0.146* | 0.984 | 0.257 | 0.956 | 0.165 |
| | | | | $Y_{42}$ | 0.252* | | 0.881 | | 0.260 | | 0.984 | |
| [24] | 1000 | 25 | 0.2 | $Y_{41}$ | 0.168 | 0.082 | 0.120* | 0.045* | 0.287 | 0.109 | 0.195 | 0.057 |
| | | | | $Y_{42}$ | 0.037* | | 0.166 | | 0.055 | | 0.290 | |
| [7] | 2400 | 19 | 1 | $Y_{41}$ | 0.313 | 0.198 | 0.183* | 0.186* | 0.391 | 0.220 | 0.213 | 0.199 |
| | | | | $Y_{42}$ | 0.110 | | 0.317 | | 0.102* | | 0.396 | |
| [38] | 3000 | 300 | 5 | $Y_{41}$ | 0.008 | 0.023 | 0.004* | 0.017* | 0.018 | 0.039 | 0.010 | 0.039 |
| | | | | $Y_{42}$ | 0.004* | | 0.009 | | 0.009 | | 0.025 | |

* least value of index

The second stage was for each expert to fill in two pairwise comparison matrices of the four cost types ($C_i$), formula (11), using scales 1–9 and 1–7. Given the reciprocal matrices, the next step was to derive priorities $w_i$ and to define C.I. according to the alternative $Y_{41}$ ([34]) and alternative $Y_{42}$ ([41]) based on priority vector methods (alternatives $Y_{32}$ and $Y_{33}$). The results of the calculations for the five supply chains are shown in Table 14, e.g. for [12] the C.I. value was 1.117 (scale 1–9, alternatives $Y_{32}$ and $Y_{41}$).

In the third step, based on the economic order quantity (EOQ) model and the dependence for calculating safety stock when lead time is constant ([5]), the actual values of the supply chain cost components $C_i$ and priorities $w_{ai}$ as components of the vector $\overline{w}_a = (w_{a1}, w_{a2}, \ldots, w_{an})$ were calculated as follows:

$$w_{ai} = C_i \Big/ C_{\sum} for\ i = 1, 2, \ldots, n \tag{12}$$

The obtained values of $w_i$ and $w_{ai}$ made it possible to determine the mean values of absolute deviations Δ, formula (9), shown in Table 14.

An analysis of Table 14 suggests the following conclusions:

- the lowest values for C.I. and Δ were obtained when the matrices were generated by experts using the scale of 1–7 (13 out of 15 estimates);
- in the considered example there is a significant variation in the values of the calculated indicators: for the mean values of absolute deviation Δ from 0.017 to 0.257, for C.I. from 0.004 (scale 1–7) to 1.208 (scale 1–9);
- only 10 out of 40 calculated values meet the C.I.≤ 0.071 limit, corresponding to the C.R.≤ 0.08 limit, and 8 of these values refer to the supply chain presented in the study of [38];
- it should be noted that a definite relationship exists between C.I. and Δ, reflecting an increase in C.I. with increasing Δ.

Thus, the ambiguity of the results underlines the need to investigate further the different alternatives presented in the morphological table (Table 2), particularly the objects that allow experts to obtain and compare qualitative and quantitative estimates.

## 6. Discussion of validating AHP

In the classical formulation of AHP, the vector $\overline{Q}$ is obtained by multiplying the pairwise comparison matrix $A$ by the priority vector $\overline{w}$:

$$\overline{Q} = A \times \overline{w}.$$

- $A$ is the pairwise comparison matrix, where each element $a_{ij}$ represents the relative judgment or importance of item $i$ compared to item $j$.
- $\overline{w}$ is the priority vector, containing the weights or ratings for each item, which are derived from the comparisons in matrix $A$.
- $\overline{Q}$ is defined as the product of matrix $A$ and the priority vector $\overline{w}$ (i.e., $\overline{Q} = A \times \overline{w}$). Each element $q_i$ in $\overline{Q}$is computed as the sum of the products of the comparisons in the corresponding row of $A$ and the weights in $\overline{w}$.

In an ideal scenario – that is, when $A$ is perfectly consistent – the following relationship holds:

$$A \times \overline{w} = \lambda_{\max}\overline{w},$$

so that each element $q_i$ in $\overline{Q}$ satisfies

$$q_i = \lambda_{\max} w_i.$$

This implies that in a perfectly consistent matrix, the ratio $w_i q_i$ is exactly constant and to $\lambda_{\max}$ for all $i$. The reade can view this constant $c$in the expression $\overline{Q} = c\overline{w}$ as the principal eigenvalue $\lambda_{\max}$.

The research present a wide range of results derived from different methodologies, scales (1–9 and 1–7), and types of objects (tangible and intangible). The authors suggest a systematically cross-compared discussion of these results. Although several tables (e.g., Tables 5, 6, 10, 12, and 14) offer detailed results on inconsistency indices, eigenvalue calculations, and deviations, the discussion tends to address these findings independently. The future studies could consider the wide approach regarding the overall picture and how the different sets of results interact and corroborate one another. The ambiguity regarding the optimal approach among alternatives $Y_{31}$– $Y_{34}$ reflects variability in the consistency indices (as noted in Table 10) and further emphasizes the necessity of cross comparison. A systematic integration or cross-comparison would help clarify under what conditions one method might outperform another. The fragmented discussion might undermine the potential to draw comprehensive conclusions regarding the reliability and practical utility of the AHP methodology. A cross comparison of results would not only validate the empirical findings (for example, the rarely used upper limits in the scale) but also support the recommendation of adopting a simpler scale (1–7) to reduce cognitive load. Therefore, the results require a more integrated discussion to highlight interdependencies and offer a clearer rationale for methodological refinements.

However, in practical applications the matrices are rarely fully consistent. In such cases, the ratios $w_i q_i$ vary around $\lambda_{\max}$, and their average is used as an estimate of the maximum eigenvalue. This variation is the basis for assessing the consistency of the judgments made in the matrix. Eqs. (2) and (4) in the manuscript reflect two different computational approaches to determine $\lambda_{\max}$ and, by extension, the consistency index (C.I.).

This process (and the associated Eqs. (2) and (4)) is central to validating the AHP because it provides a metric for how closely the matrix aligns with the theoretical expectation. A higher degree of deviation indicates a higher level of inconsistency in the judgments, suggesting that expert assessments should be reviewed.

## 7. Conclusion and future research

One of the possible ways to summarize the results of studies devoted to AHP is the method of morphological analysis and synthesis. The developed option of the morphological table includes eight basic features, each containing two to five alternatives (Table 2). It has been established that the number of options for forming AHP is 4860 (without considering the correlation between the alternatives).

The developed structural model $S(Y_{ij})$, which covers all computational operations of AHP, consists of two interrelated models. The first model includes four features of the morphological table and is designed to calculate the priority vector $w$, maximum eigenvalue $\lambda_{max}$ and consistency index C.I. The second model, uniting the control and adjusting operations, provides for a calculation of the random consistency index R.I., consistency ratio C.R., assessment of the correctness of the results and, if necessary, revision of judgments. This paper pays primary attention to the first model, which is the basis of AHP.

Based on analyzing studies by Saaty ([34,35]) that are the basis for forming AHP, the authors assess matrices reflecting judgments of experts and the inevitable results of their processing, considering the specifics of tangible or intangible objects of assessment and the number of consistent and inconsistent matrices. The results presented in Tables 3 and 4 contradict one of Saaty's primary goals: to demonstrate the possibility of relative measurement of intangible objects by comparing them with tangible ones.

A statistical analysis of almost 500 matrices (Tables 7 and 8) has shown, first, that the upper triangular section of the matrices is filled by about 70 % with numbers 1, 2, …, 9 and by 30 % by the reciprocals of those numbers (1/2, 1/3, …, 1/9). Second, it has shown the empirical probability of using numbers 8 and 9 is $P(8+9) = 0.12$, and third, that the obtained data contradict Saaty's basic hypothesis of the uniform

distribution of numbers 1–9 in simulating the random consistency index R.I., according to which the probability of using numbers 8 and 9 is $P(8+9)=0.24$. Thus, the simulation algorithm R.I. (feature $H_5$ of the morphological table) should be revised. The results are significant: the R.I. value is used to calculate the consistency ratio C.R.=C.I./R.I., with most studies (such as those shown in Table 1) indicating the need to revise expert judgments when C.R.> 0.1. The results also allow to expand the decision makers with the application of the coefficient of concordance of experts' opinions. For this purpose there is in the work all data additionally to conduct nonparametric statistical test Kendall's Concordance.

Despite the contributions of this study to advancing the AHP methodology, several limitations must be acknowledged. First, although an extensive set of nearly 500 expert judgment matrices was analyzed, the range of tangible and intangible objects remains somewhat limited. This sample size, while substantial, may not fully capture the variability in expert behavior across diverse decision contexts. In particular, the empirical evidence indicates that experts rarely utilize the upper extremes of Saaty's 1–9 scale, prompting the recommendation for a simplified 1–7 scale. However, the impact of this scale reduction on more complex or larger matrices warrants further investigation. Moreover, the study identifies inconsistencies among the various methods for deriving priority vectors. The lack of a clearly superior method, as seen in the balanced performance among alternatives ($Y_{32}$, $Y_{33}$, and $Y_{34}$) in certain scenarios, suggests that the traditional computational framework may not be sensitive enough to different contexts. The division between tangible and intangible assessment further complicates the issue—calculations of deviation indices ($\sigma$ and $\Delta$) are not always feasible for intangible objects, rendering cross-comparison incomplete. Another limitation arises from the simulation algorithm used to compute the random consistency index (R.I.). Empirical probabilities, particularly for the use of numbers 8 and 9, contradict the uniform distribution assumptions underlying the traditional AHP. This discrepancy indicates a need for a revised R.I. simulation procedure that better reflects actual expert usage patterns.

In light of these limitations, future research should aim to:

1. Expand the Empirical Base: Increase the number of matrices, especially for tangible objects, to ensure that the findings are robust across a wider range of decision contexts.

2. Refine Computational Methods: Conduct further comparative analyses of priority vector derivation techniques. Future studies should explore context-sensitive, hybrid approaches that integrate probabilistic-statistical dependencies to enhance the reliability of the derived indices.

3. Revise the Simulation Algorithm: Address the discrepancies in the R.I. computation by developing an updated simulation algorithm that accounts for the empirical non-uniformity in expert judgments.

4. Integrate Dynamic Feedback: Investigate mechanisms for incorporating feedback loops in the AHP framework, particularly in the iterative revision of expert judgments where consistency ratios exceed acceptable thresholds.

5. Examine Scale Implications: Further analyze the implications of adopting the simplified 1–7 scale versus the traditional 1–9 scale across various matrix dimensions and decision settings. This would include sensitivity analyses to understand how scale modifications affect overall consistency and reliability.

Addressing these avenues would not only refine the AHP methodology theoretically but also enhance its practical utility in complex, real-world decision-making scenarios.

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## CRediT authorship contribution statement

**Bazhina Darya:** Validation, Supervision, Software, Resources. **Lukinskiy Vladislav:** Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation. **Lukinskiy Valery:** Writing – original draft, Project administration, Conceptualization.

## Declaration of Competing Interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

## Data availability

The data used to support the findings of this study are available from the corresponding author upon request. Any other data are available from the corresponding author upon request.

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